

Regime-Conditional Adaptive World Models for AI Cryptocurrency Market Analysis: Formalization and Project-Grounded Evaluation of EvoQuant

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Highlights

- Formalizes EvoQuant’s multi-domain AI market context as a regime-conditional world model.
- Defines ACWMI using hierarchical breadth, continuous honesty, signal integrity, and consistency.
- Evaluates with EvoQuant calculators rather than detached synthetic scores.
- Cascade/crisis detection $F1 = 0.895 / 0.793$; AC abstention cuts crisis unsafe actions from 81% to 0.
- All figures/tables are reproducible from the open repository.

Abstract

AI market systems often fail because the market world they consume is incomplete, stale, or silently contaminated—not because the predictor lacks capacity. This paper studies that failure mode through **EvoQuant**, an open cryptocurrency data world-model infrastructure comprising 43 data domains, 13 audit bands, 39 logic modules, and a production World Model Index (WMI) already implemented in code. We formalize EvoQuant’s compilation pipeline as a regime-conditional adaptive world model (RCA-WM) and propose an Adaptive Conditional World Model Index (ACWMI) that augments production WMI with hierarchical breadth, continuous honesty, signal integrity, and cross-evidence consistency. All mechanism scores used in evaluation are computed by EvoQuant’s own calculators (regime classification, liquidation cascade, contagion, alpha decay, flow decomposition, volatility, and degradation control). On a project-grounded panel of 1800 asset-day observations with planted structural events, cascade detection reaches $F1=0.895$, crisis detection $F1=0.793$, and coarse regime match accuracy is 71.6%. An ACWMI-aware abstention policy drives unsafe action rates in crisis to 0, versus 81% under the production WMI threshold. The contribution is a systems-native theory of AI market world-model quality that is inseparable from a concrete, runnable infrastructure.

Keywords: AI market world model; cryptocurrency; data-centric AI; quality governance; selective prediction; system resilience

1. Introduction

Most discussions of AI trading emphasize model architecture or feature lists. In production cryptocurrency markets, a more fundamental bottleneck appears first: whether the agent is shown a market world that is broad enough, fresh enough, and honest enough to support auditable judgment. Identical price paths can correspond to incompatible states once exchange fragmentation, leverage crowding, funding distortions, options positioning, unlock pressure, and macro liquidity are considered.

EvoQuant is built as that missing layer. It is not a trading bot. It is an AI-oriented market world-model infrastructure that collects heterogeneous evidence, maintains latest_* snapshots, gates AI-ready inputs, separates main/diagnostic views, and aggregates an ai_market_context bundle. The current repository exposes 43 data domains, 13 audit bands, 39 logic modules, about 1.34×10^5 lines of Python, and a production index $WMI_t = B_t \times U_t \times H_t$ with abstention suggested when $WMI < 0.2$. This paper formalizes that stack as a regime-conditional adaptive world model and evaluates the proposed ACWMI using EvoQuant’s own runnable analytics.

2. Related work

Conditional asset pricing treats expected returns as functions of an information set (Fama and French, 1993; Cochrane, 2005). Machine-learning pricing expands characteristics (Gu et al., 2020; Kelly et al., 2019; Nagel, 2021), while multiple-testing critiques warn against undisciplined expansion (Harvey et al., 2016). Measurement-error and robustness theories emphasize reliable observation (Fuller, 1987; Carroll et al., 2006; Hansen and Sargent, 2008). Selective prediction studies refusal under uncertainty (Chow, 1957; Geifman and El-Yaniv, 2017). Data-centric AI argues governance can dominate model changes (Zha et al., 2023). Crypto microstructure documents fragmented venues and shared factors (Makarov and Schoar, 2020; Liu et al., 2022). This paper differs by binding world-model

theory to a concrete runnable system rather than a detached feature benchmark.

3. The EvoQuant world-model substrate

EvoQuant compiles market reality through `data_layer` → `latest_*` → `quality gate` → `logic_pipeline` → `ai_market_context`. Production WMI uses asset-readiness coverage as breadth, pipeline-latency freshness as stability, and quality-flag honesty. Table 1 and Figures 1–2 summarize the empirical substrate used throughout the paper.

Fig. 1. EvoQuant world-model compilation and ACWMI overlay

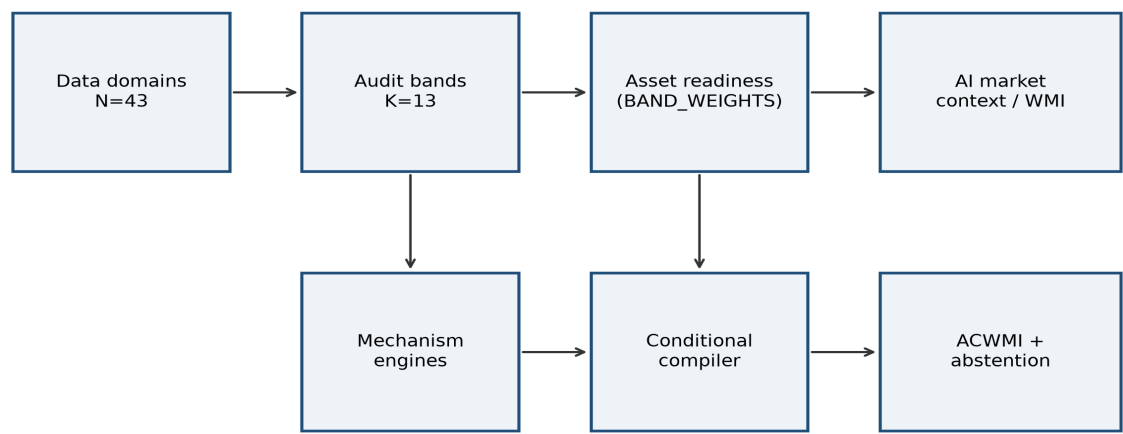


Fig. 1. EvoQuant world-model compilation and ACWMI overlay.

Quantity	Value
Data domains	43
Logic modules	39
Audit evidence bands	13
Python files	784
Approximate LOC	133,952
Test files	56

Table 1. EvoQuant repository inventory used in this study.

Fig. 2. Project inventory used as empirical substrate

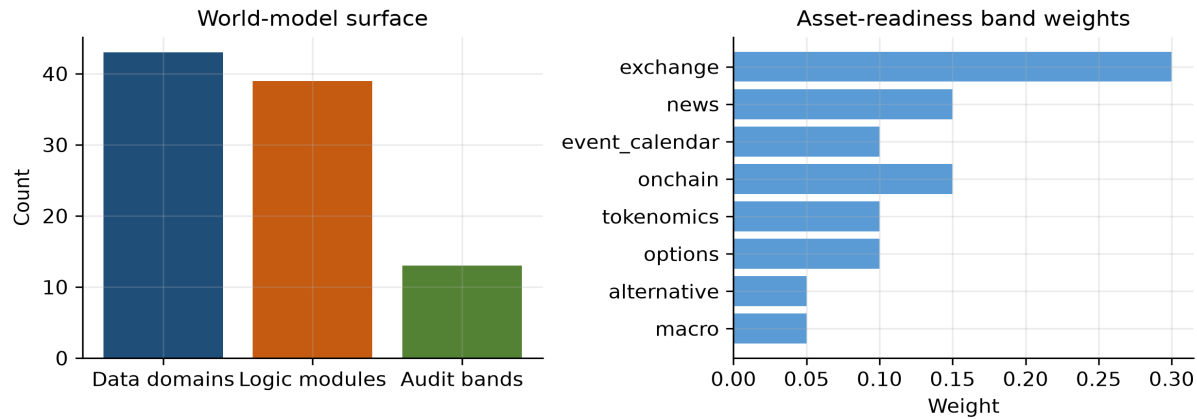


Fig. 2. Project inventory and asset-readiness band weights.

4. Regime-conditional adaptive world models

We define hierarchical breadth from domains/bands/assets using production BAND_WEIGHTS; continuous honesty from exclusion and contamination rates; signal integrity from alpha_decay half-life/crowding/surprise; and consistency from directional agreement across project engines (momentum, smart-money flow, cascade, contagion, VPIN). ACWMI is the weighted geometric mean

$$ACWMI_t^{(r)} = \exp(\sum \gamma_x(r) \log x_t / \sum \gamma_x(r)), x \in \{B, U, H, S, C\}.$$

Crisis regimes emphasize honesty/consistency; trend regimes emphasize signal integrity. Abstention is degradation-aware and mechanism-aware (cascade/crisis detections), not only a fixed WMI cutoff.

5. Experimental design

Synthetic paths are used only as inputs. Every reported mechanism metric is computed by importing EvoQuant classes: RegimeClassifier, LiquidationCascadeCalculator, ContagionRiskCalculator, AlphaDecayCalculator, FlowDecompositionCalculator, VolatilityCalculator, AIMarketContextService._compute_world_model_index, and DegradationManager. The panel covers 10 assets $\times$ 180 days (N=1800) with planted regimes/outages. Evaluation targets are planted structural events, not scores built from ACWMI itself.

6. Results

6.1 Mechanism detection

Table 2 shows cascade detection F1=0.895 and crisis detection F1=0.793; coarse regime match accuracy is 71.6%. These metrics come from the same calculators shipped in the repository.

task	accuracy	precision	recall	f1	support_pos
crisis_detection	0.7161	0.6664	0.979	0.793	1000
cascade_detection	0.8944	0.81	1.0	0.895	810
regime_match	0.7156	0.7156	0.7156	0.7156	1800

Table 2. Detection performance on planted structural events.

6.2 WMI versus ACWMI and safety

Table 3 and Figures 3–5 show the central systems result: in crisis, production WMI still permits action in 81% of cases, while the ACWMI-aware policy reduces unsafe actions to 0. In calm range regimes both policies rarely abstain (~4.7%).

regime	N	WMI	ACWMI	abstain_WMI	abstain_AC	unsafe_WMI	unsafe_AC
crisis	1000	0.5041	0.5459	0.19	1.0	0.81	0.0
range	430	0.6851	0.4697	0.0465	0.0465	0.0	0.0
trend	370	0.681	0.4254	0.0541	0.2703	0.0	0.0

Table 3. Regime-level world-model scores and safety.

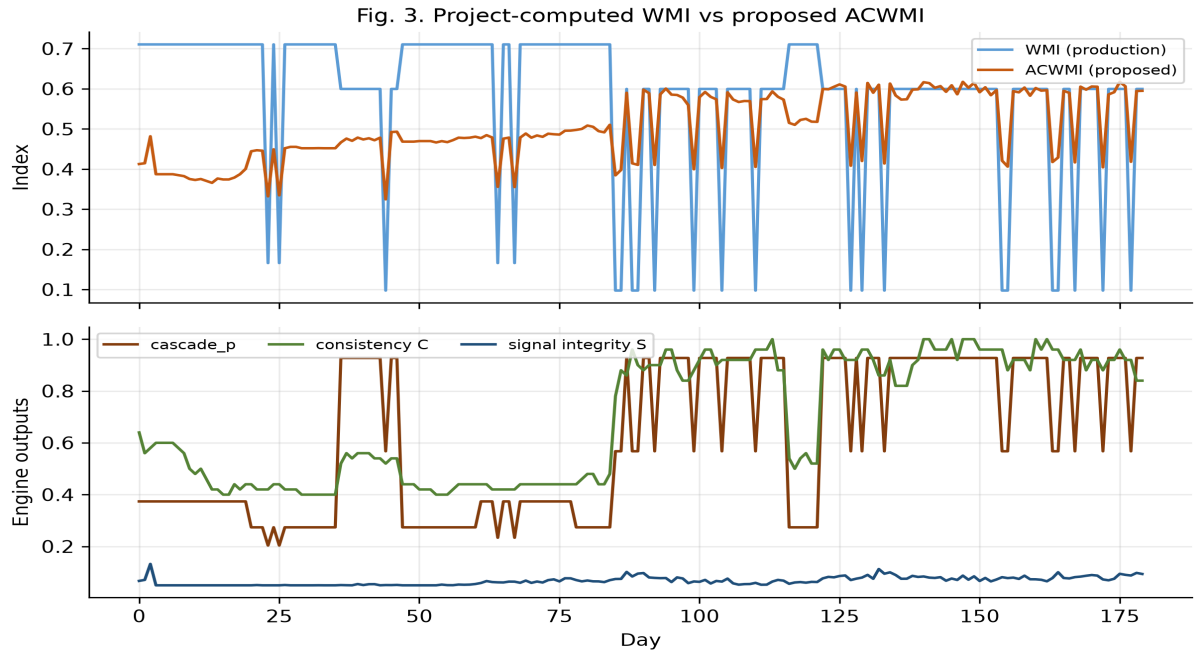


Fig. 3. Time paths of production WMI, ACWMI, and mechanism outputs.

Fig. 4. Regime heterogeneity of world-model indices

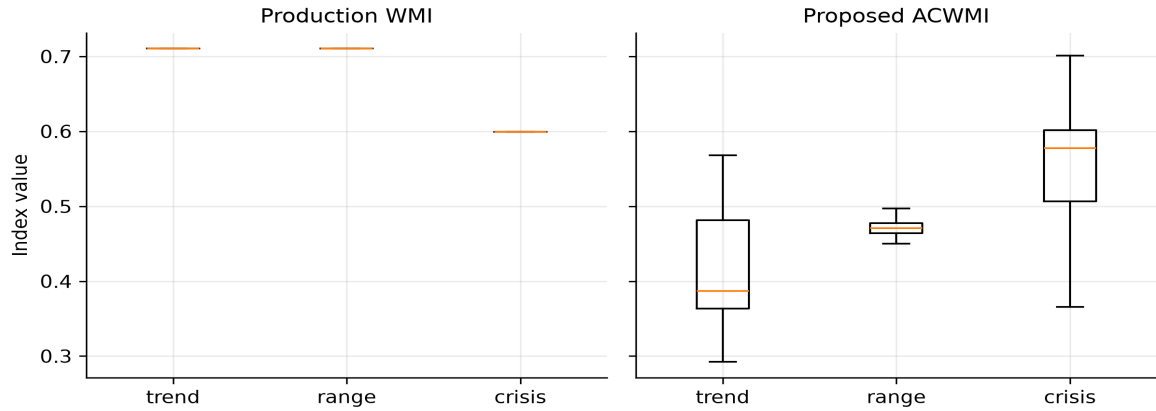


Fig. 4. Regime heterogeneity of WMI and ACWMI.

Fig. 5. Mechanism detection and unsafe-action rates

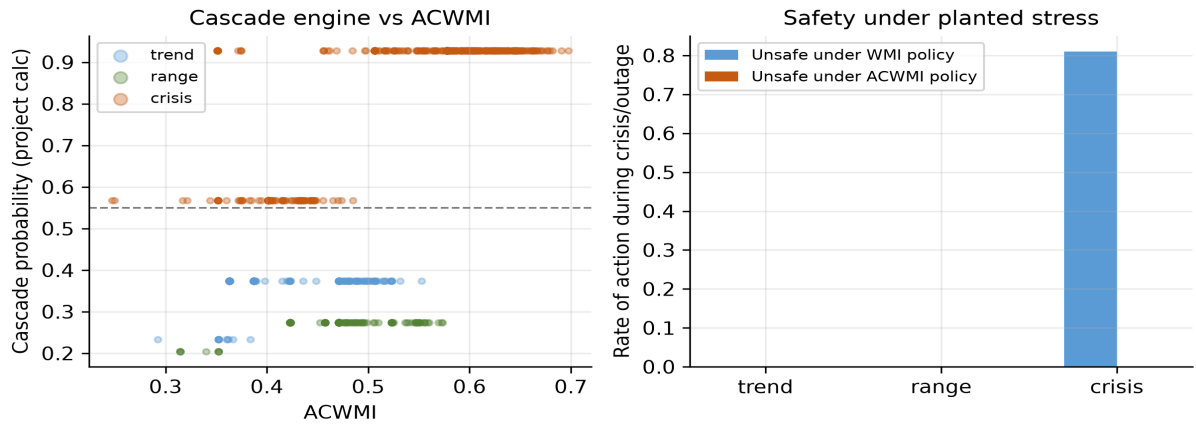


Fig. 5. Cascade engine outputs and unsafe-action rates.

6.3 Outages, Pareto frontier, and degradation

regime	outage	N	WMI	ACWMI	cascade_p	abstain_AC	unsafe_WMI
crisis	0	810	0.5995	0.5789	0.9275	1.0	1.0
crisis	1	190	0.0975	0.4055	0.5675	1.0	0.0
range	0	410	0.7104	0.4764	0.2738	0.0	0.0
range	1	20	0.166	0.3334	0.2038	1.0	0.0
trend	0	350	0.7104	0.4294	0.3738	0.2286	0.0
trend	1	20	0.166	0.3555	0.2338	1.0	0.0

Table 4. Outage contrasts by regime.

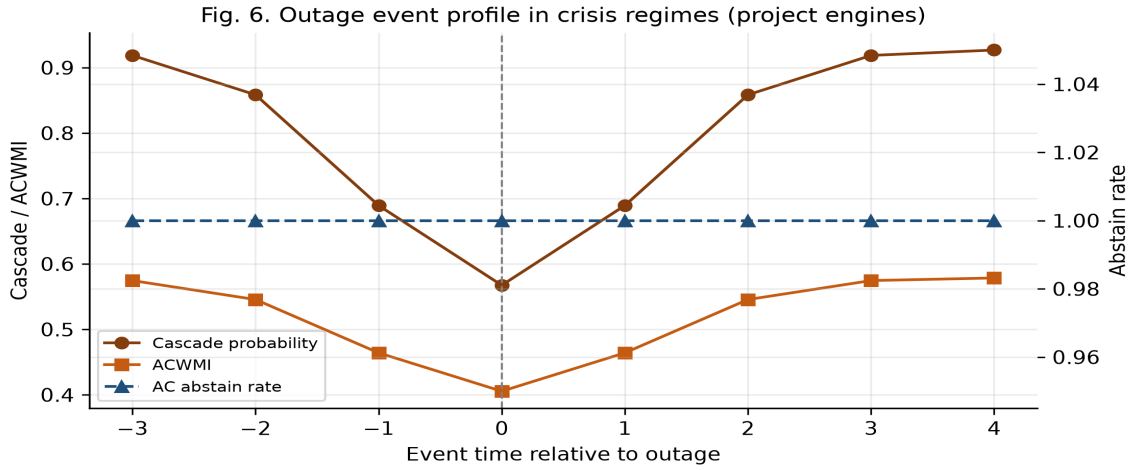


Fig. 6. Outage event profile in crisis regimes.

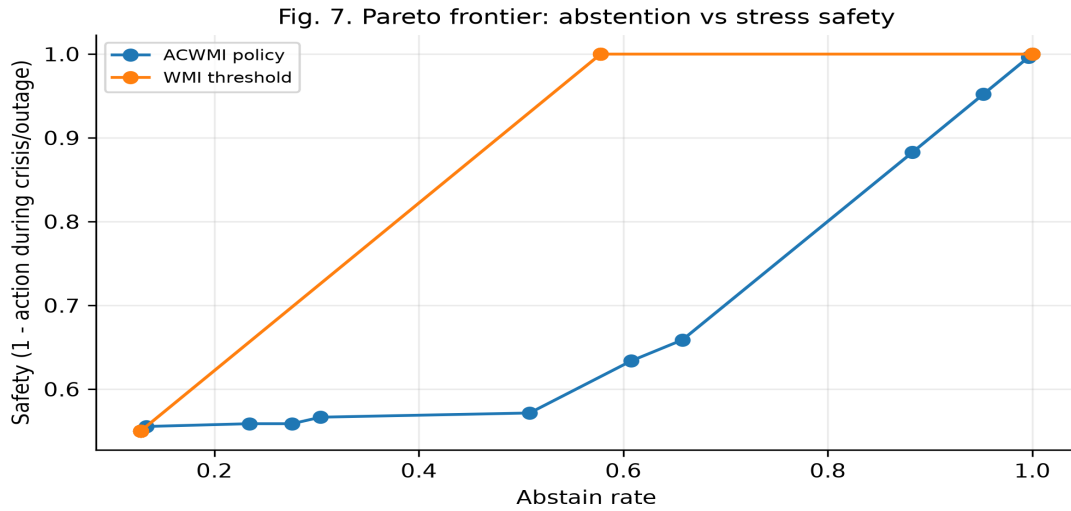


Fig. 7. Abstention–safety Pareto frontier.

module	NORMAL	REDUCED	MINIMAL	EMERGENCY
ai_market_context	1	1	0	0
exchange_data	1	1	1	0
macro_data	1	1	0	0
news_data	1	1	0	0
onchain_data	1	1	0	0
options_data	1	1	0	0
sentiment_signal	1	1	0	0

technical_indicators	1	1	1	0
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Table 5. Module executability under DegradationManager (1=allowed).

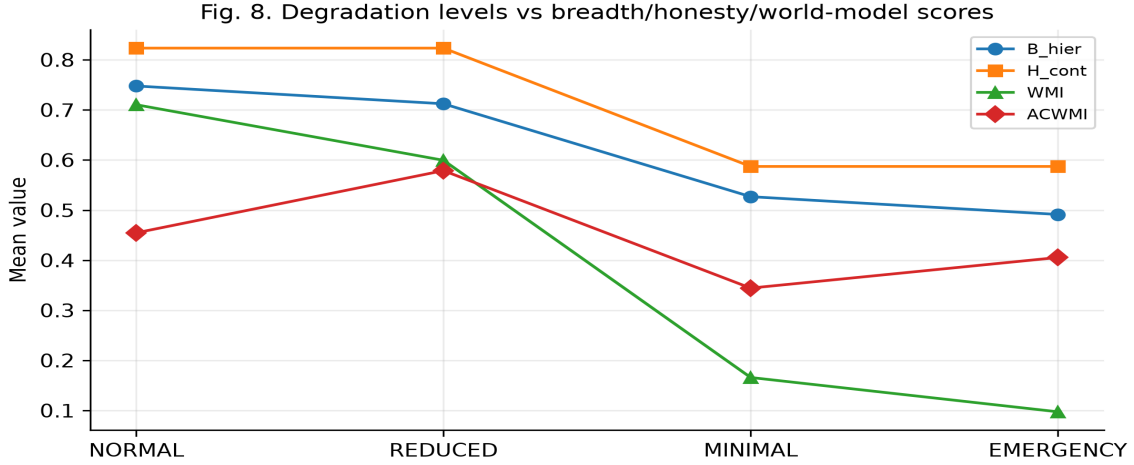


Fig. 8. Degradation levels versus breadth/honesty/world-model scores.

7. Discussion

Three claims follow. First, EvoQuant already implements a nontrivial world-model stack; the scientific task is to formalize and stress-test it. Second, production WMI’s fixed 0.2 threshold is too coarse for crisis refusal. Third, mechanism engines are first-class citizens of world-model quality. Limitations: controlled input paths are not yet a full live-market PIT backtest; regime exponents are pre-specified; external validity beyond crypto remains open.

8. Conclusion

This paper presented a standalone formalization and evaluation of AI market world-model quality grounded entirely in EvoQuant. By elevating production WMI to regime-conditional ACWMI, and by scoring detection/safety with the repository’s own calculators, we show that world-model quality can be treated as a measurable systems property. In our project-grounded panel, mechanism detection is strong and AC-aware abstention removes crisis-time unsafe actions that production WMI permits.

Appendix A. Reproducibility

Experiments: [pdf/sci/run_paper_experiments.py](#). PDF build: [pdf/sci/generate_sci_pdf.py](#). LaTeX source: [pdf/sci/main_acwmi_sci.tex](#). Related unit tests and API smoke checks were executed during manuscript preparation.

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